



Suparnički napadi na modele za klasifikaciju slika

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SUPARNIČKI NAPADI

- Cilj rada bio je pokazati kako dodavanje male perturbacije na ulaz može drastično utjecati na točnost klasifikacijskih modela

$$x' = x + \delta$$

- Različiti napadi imaju različite strategije za odabir perturbacije
- FGSM

FGSM

$$\mathcal{L}(\theta, x + \delta, y) \approx \mathcal{L}(\theta, x, y) + \delta^T \nabla_x \mathcal{L}(\theta, x, y)$$

$$\|\delta\|_{\infty} \leq \epsilon$$

$$\delta = \epsilon \cdot \text{sign}(\nabla_x \mathcal{L}(\theta, x, y))$$

$$x_{\text{adv}} = x + \epsilon \cdot \text{sign}(\nabla_x \mathcal{L}(\theta, x, y))$$

- Black-box napad

Ciljani napad

$$x_{\text{adv}} = x - \epsilon \cdot \text{sign}(\nabla_x \mathcal{L}(\theta, x, y_{\text{target}}))$$

Generiranje suparničkih primjera

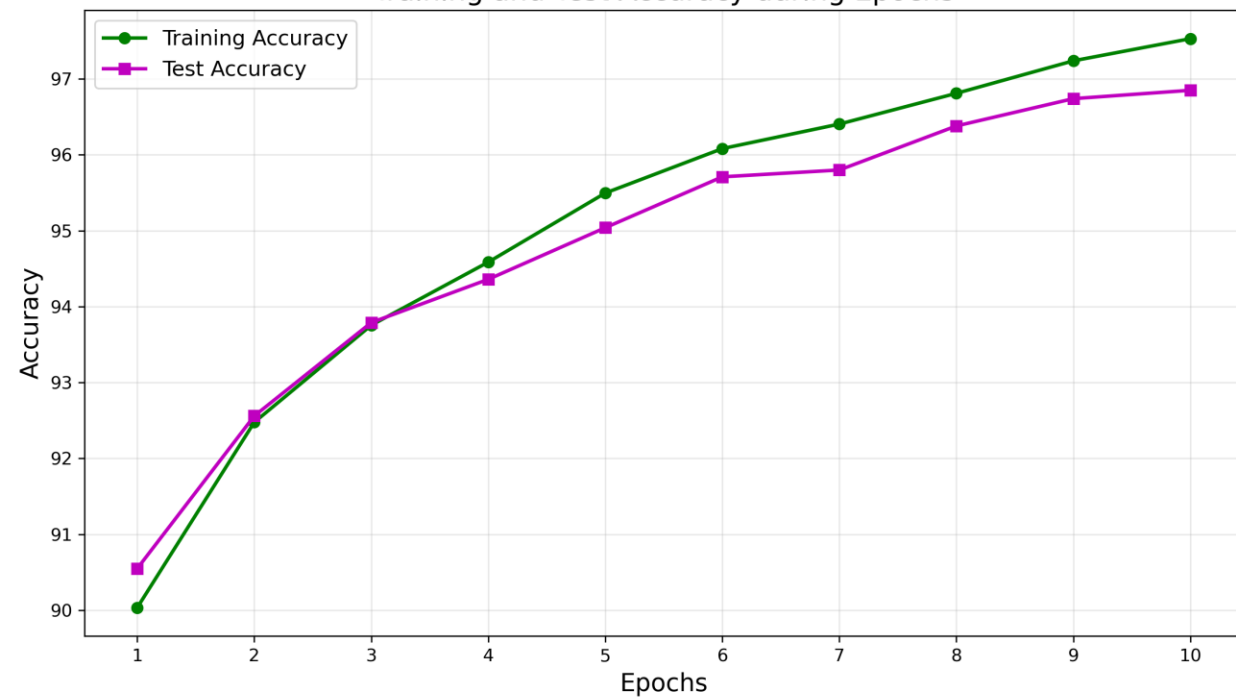
- Dodavanje perturbacije na slike iz ispitnog skupa
- Samo ako se slika inicijalno točno klasificira

Potpuno povezani model za klasifikaciju skupa MNIST

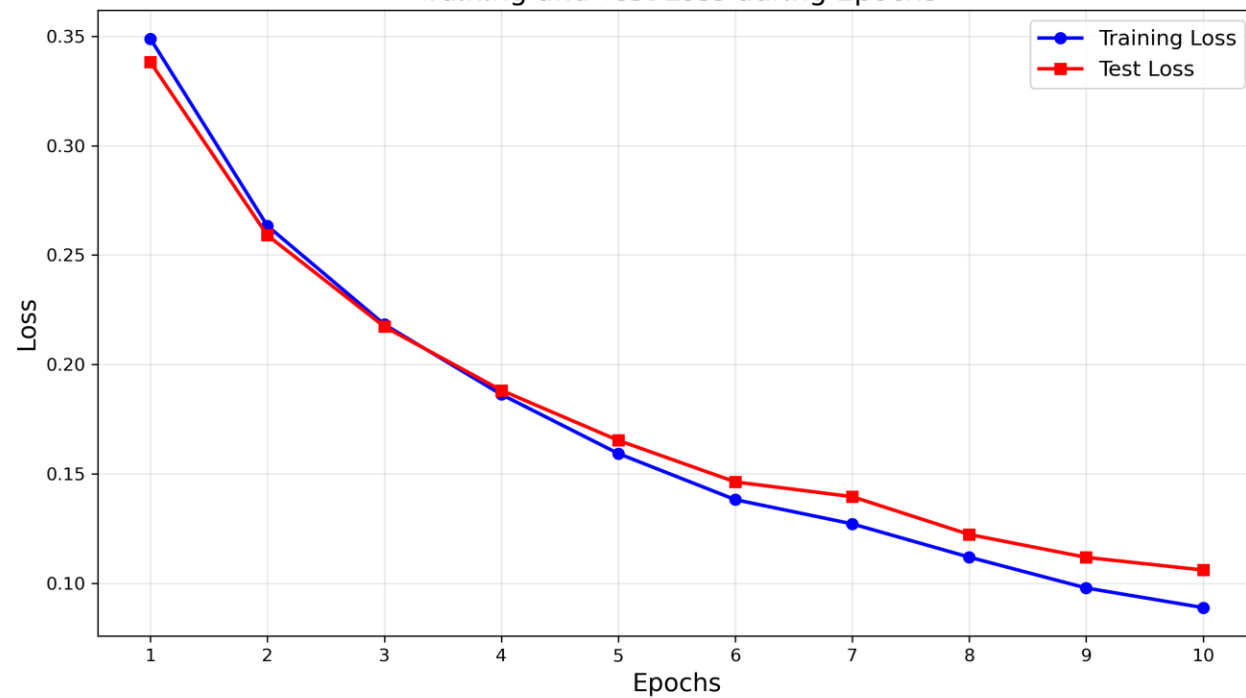
- sgd, 10 epoha
- 784x512x64x10
- ReLu, LogSoftmax
- NLL
- 70 000 jednokanalnih slika 28x28
- 10 klasa (znamenke)

Točnost i gubitak

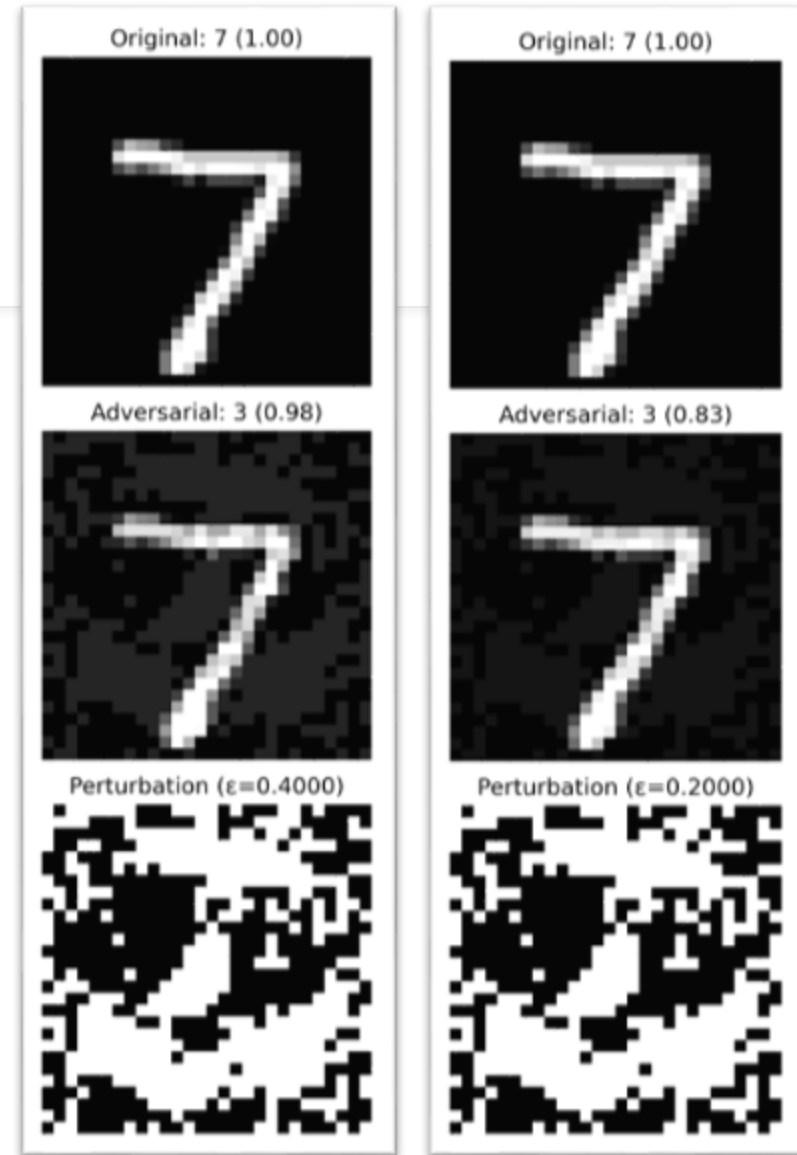
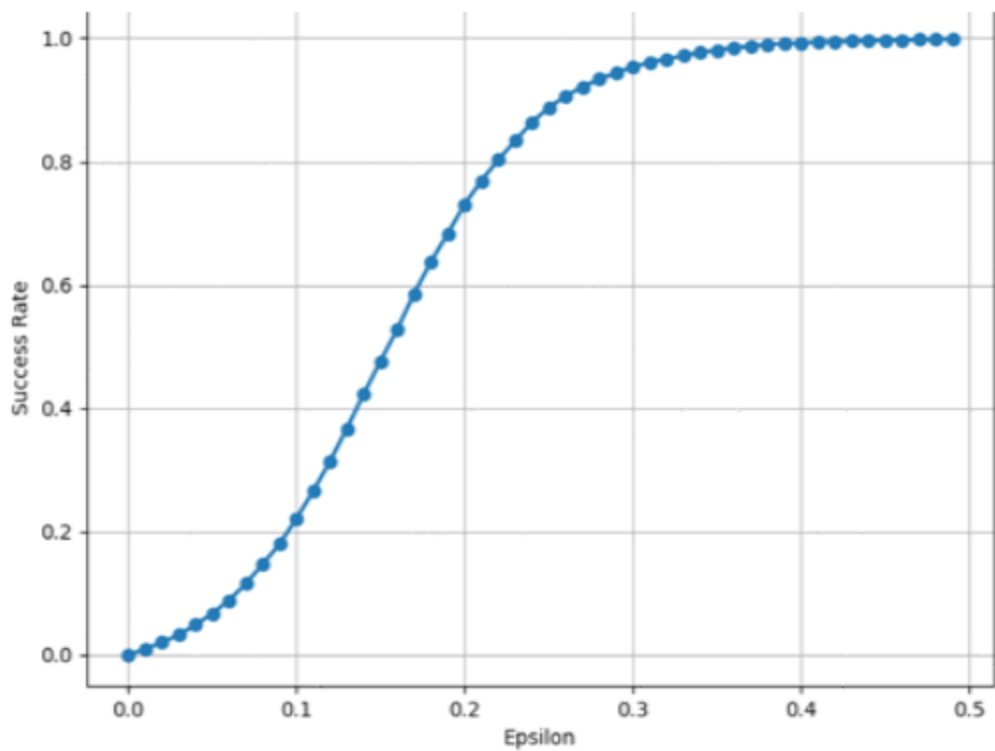
Training and Test Accuracy during Epochs



Training and Test Loss during Epochs

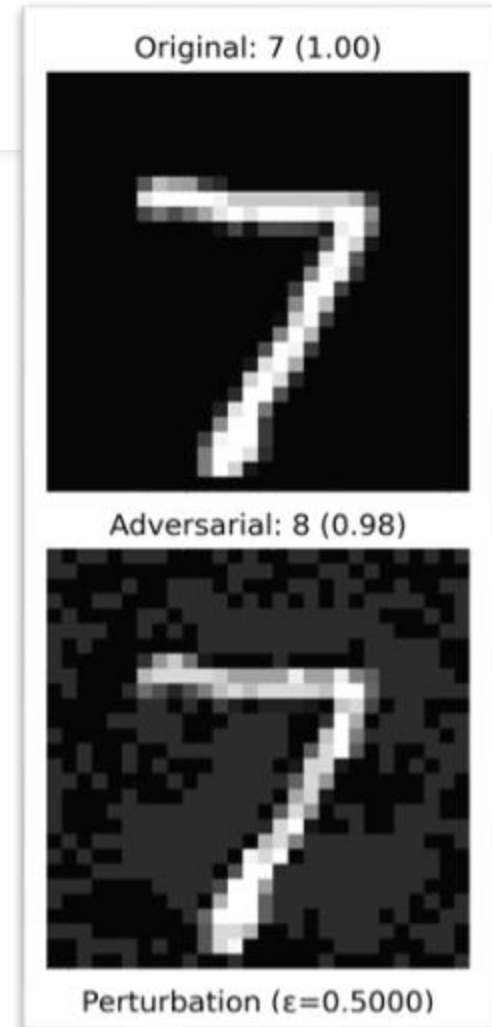
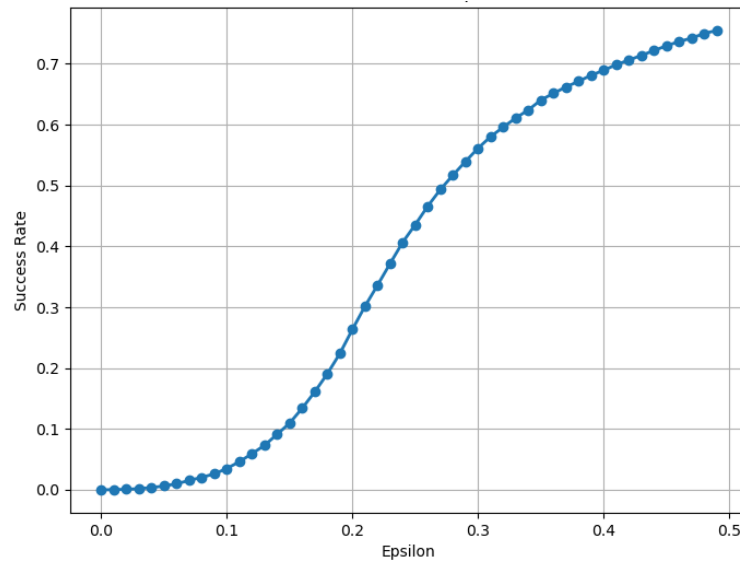


MNIST, neciljani napad



MNIST, ciljani napad

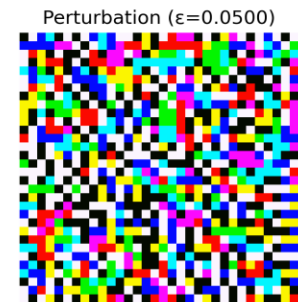
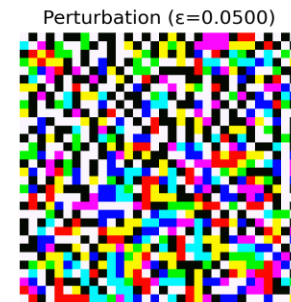
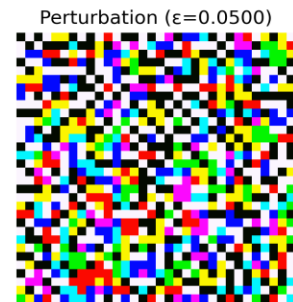
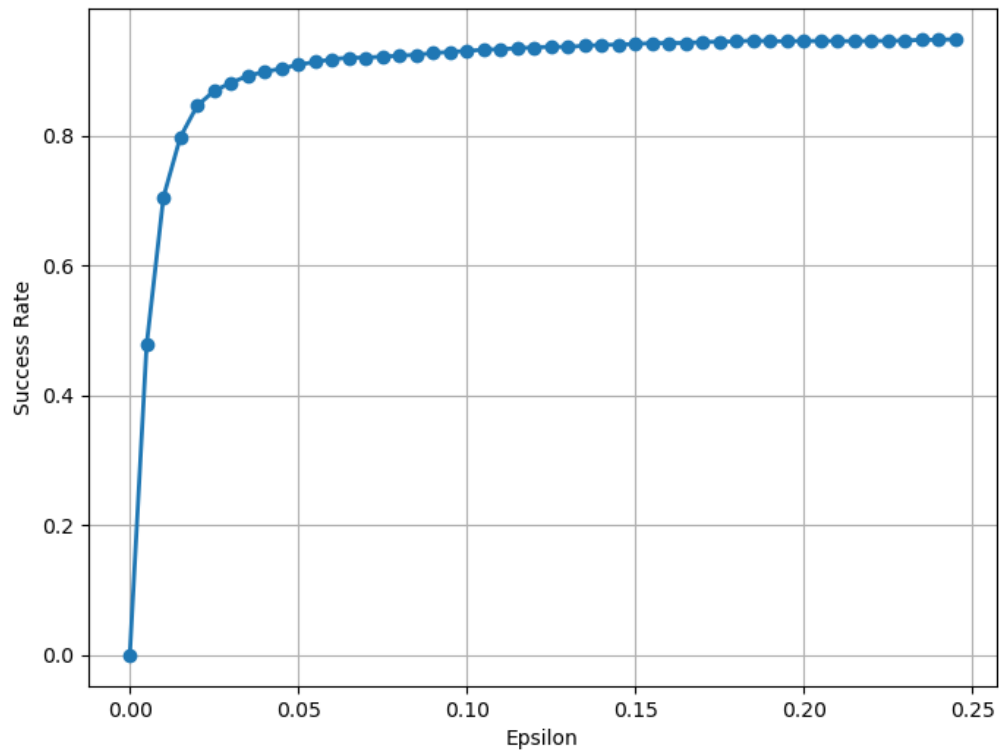
- Ciljna klasa za klasu t je klasa $t+1$



Resnet-20 za klasifikaciju skupa CIFAR-100

- Predtrenirani model
- točnost 68.83% na CIFAR-100 ispitnom skupu
- 20 slojeva, rezidualne veze
- 60 000 trokanalnih slika (RGB) 32x32
- 100 klasa podijeljenih u 20 nadklasa

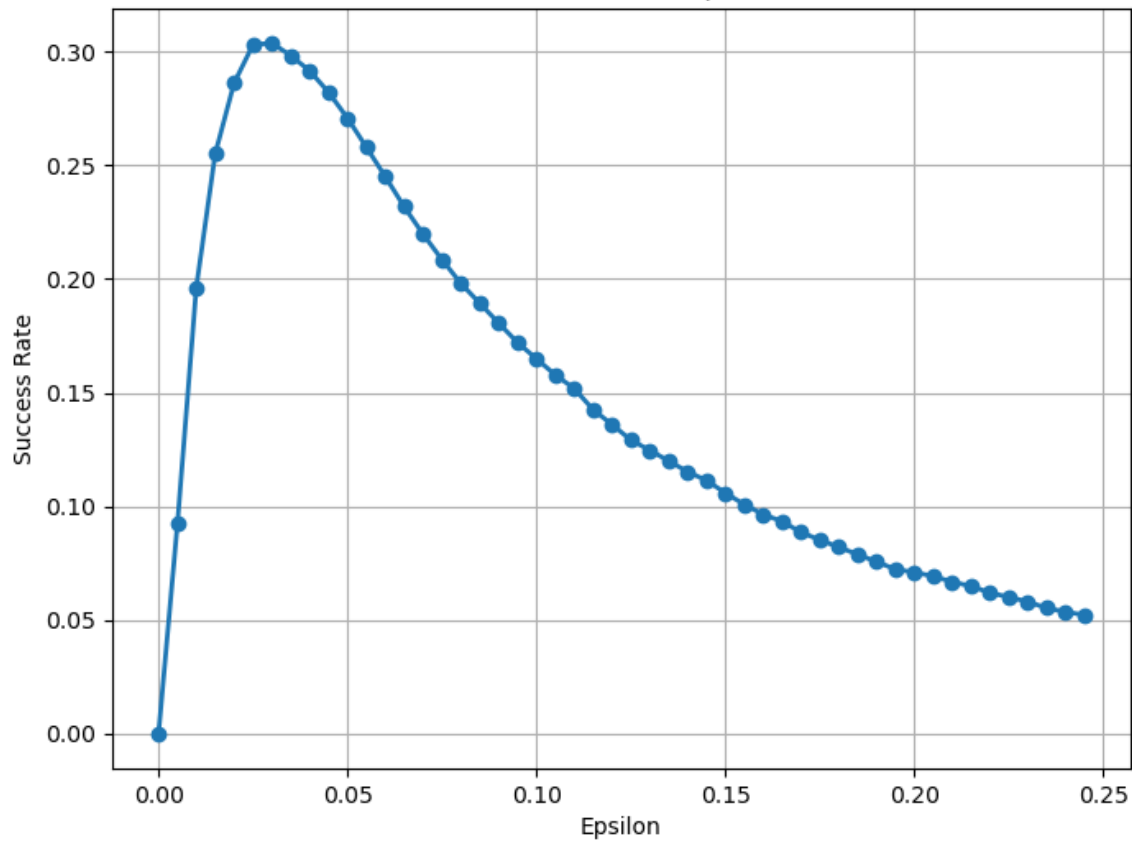
CIFAR-100 Neciljani napad



CIFAR-100, Ciljani napad

- Hijerarhijska struktura
- Napad kada je ciljna klasa u istoj nadklasi kao inicijalna klasa
- Napad kada ciljna klasa nije u istoj nadklasi kao inicijalna klasa

Ciljna klasa u istoj nadklasi



Original: seal (0.57)



Adversarial: beaver (0.98)



Perturbation ($\epsilon=0.0300$)



Original: camel (0.71)



Adversarial: cattle (0.99)



Perturbation ($\epsilon=0.0300$)



Original: lion (1.00)



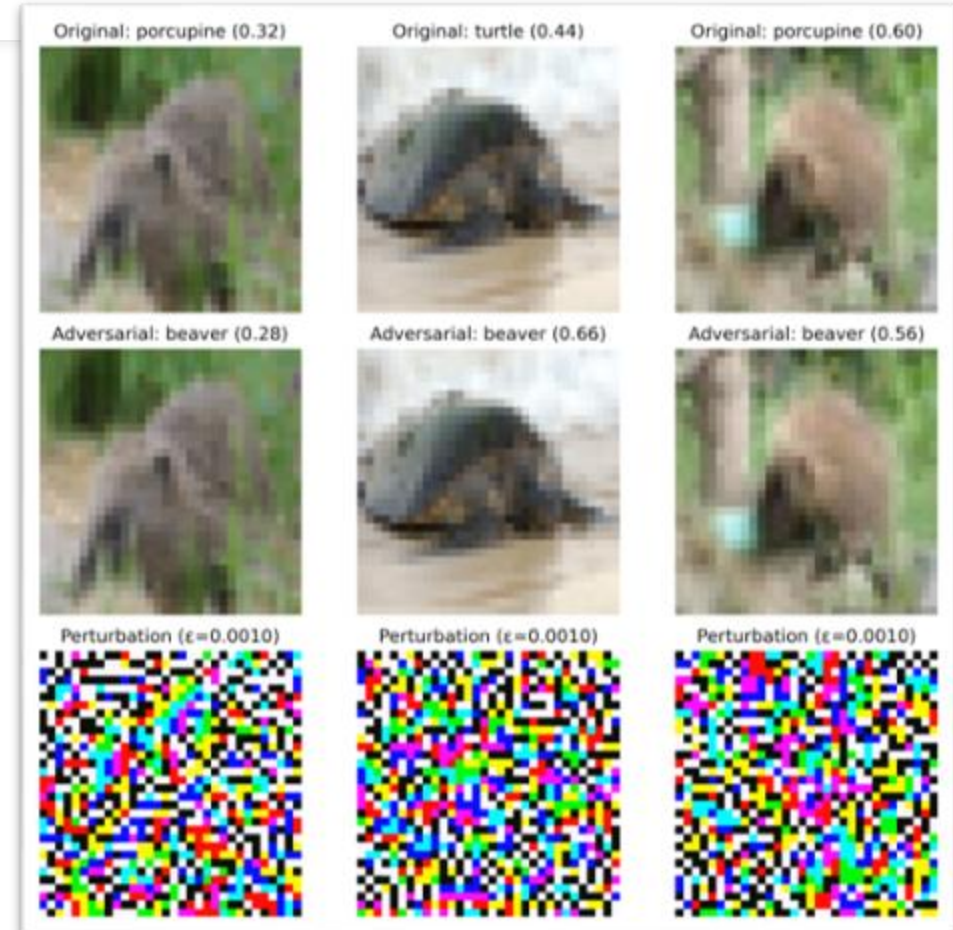
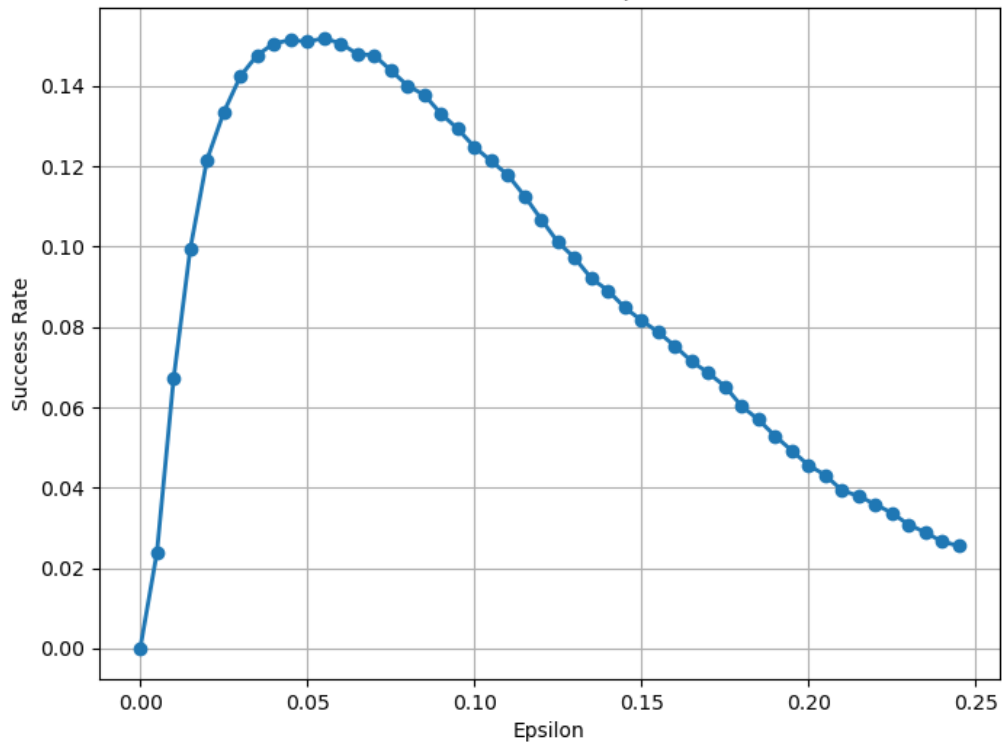
Adversarial: bear (0.39)



Perturbation ($\epsilon=0.0300$)



Ciljna klasa u različitoj nadklasi



Robusnost

- Lakše je napraviti uspješan neciljani napad na modelu Resnet-20
- Uspješnost ciljanog napada je veća za potpuno povezani model

Zaključak i budući rad

- Rad pokazuje ranjivosti modela koji postižu dobre rezultate na ispitnim skupovima na suparničke primjere
- Budući rad bi mogao uključivati:
 - Suparničko treniranje
 - Implementacija novih napada (PGD)
 - Primjena FGSM-a na nove modele



Hvala na pozornosti!